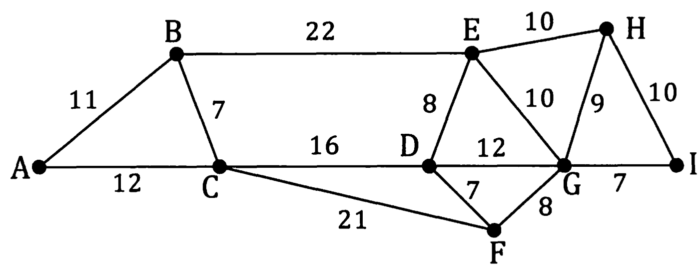
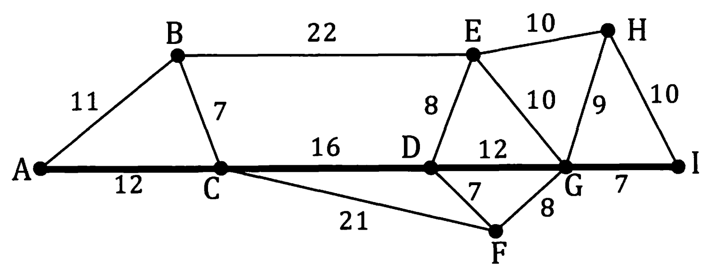
Chapter 5.
Minimum spanning trees.

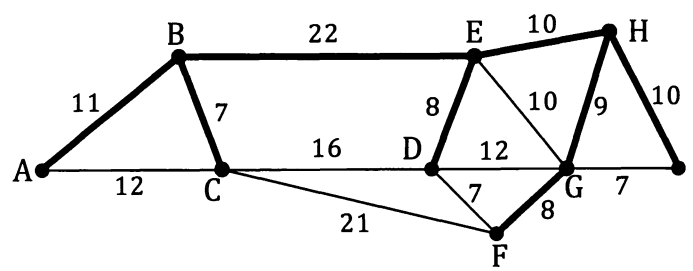
Spanning trees.
The network on the right shows the roads that exist between nine farms, A to I, with the numbers indicating distances in kilometres. The water authority wish to connect these farms to the mains water supply by laying water pipes alongside existing roads.



This is not the same as finding the shortest route from A to I because this shortest route, shown on the right, does not connect up all the farms (B, E, F and H are not "on line").



Instead we need to find a system that includes all the farms but which has no unnecessary connections. What we need is a **minimum spanning tree** for this network.

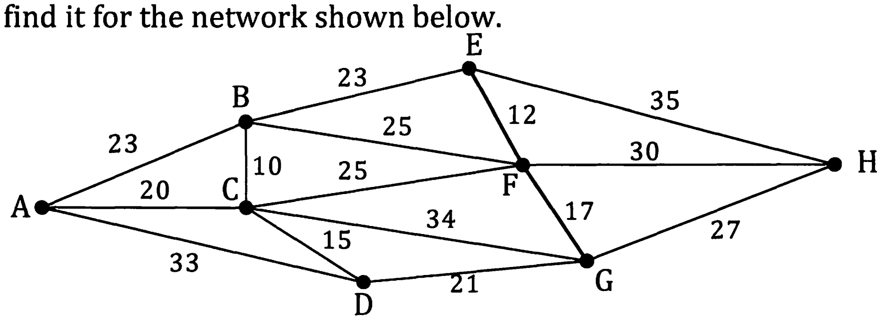


(A spanning tree of a graph is a subgraph that connects all of the vertices and is itself a tree.)

The spanning tree shown requires a total length of 85 km (= 11 km + 7 km + 22 km + 8 km + 10 km + 9 km + 10 km + 8 km).

Can you find a shorter spanning tree?
Can you find the shortest spanning tree?
How long is it?

If you think you have found the minimum spanning tree for the network shown above then try to find it for the network shown below.

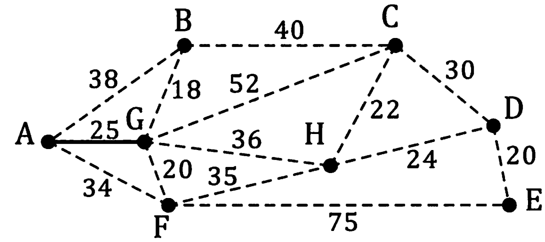


Minimum spanning trees – a systematic approach.

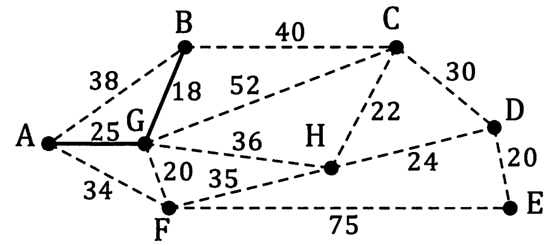
Consider the network on the right.

To determine the minimum spanning tree systematically we can in fact start at any vertex but we will start at A as that seems a logical place to start.

From A we next bring "on line" the vertex that is A's *nearest neighbour*. Point G, being just 25 units from A is the nearest neighbour in this example.

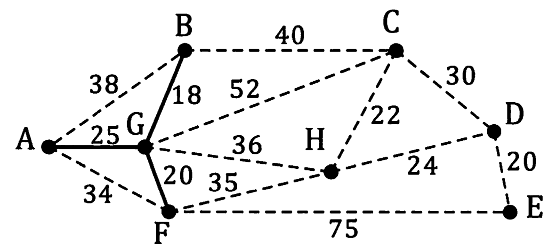


Now that we have A and G on line we next connect to the vertex that is closest to one of these two vertices. In our example B is the next nearest neighbour to one of our on line points A and G.

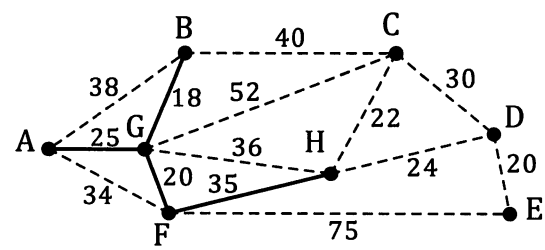


Now that we have A, G and B on line we now look for the vertex that is not yet on line but that is the nearest neighbour to one of our on line points.

Thus point F, being just 20 units from G, is the next nearest neighbour.

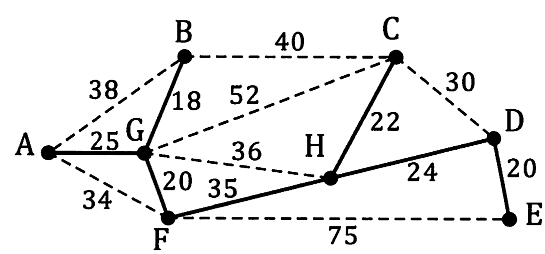


We now have A, G, B and F on line and again look for the next nearest neighbour. In this case point A is only 34 units from F but we already have A on line so this connection would be pointless. (It would give an unwanted *cycle* in our connection.) Thus we choose to bring H on line as it is the nearest neighbour not yet on line.

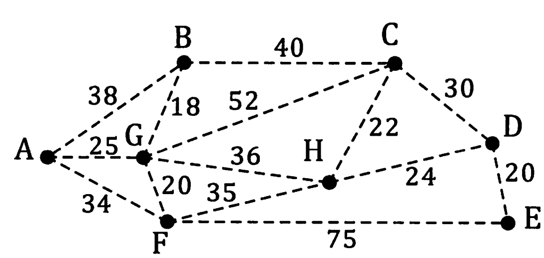


Continuing in this way until all points are on line gives the minimum spanning tree shown on the right.

Of course this process would usually be carried out on one copy of the network. The five copies used here are purely to illustrate the steps of the method.



The reader should now use this method on the copy of the network shown on the right but this time start at a vertex other than A and confirm that a different starting point still leads to the same minimum spanning tree.



The method explained on the previous page, in which we:

choose any vertex as our initial "on line" vertex and then build up the spanning tree by connecting online vertices to the "nearest neighbour", whilst always making sure that no cycles are introduced,

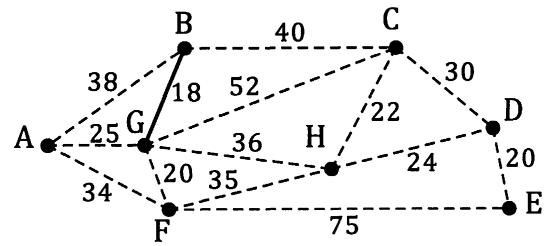
is called **Prim's algorithm**.

An alternative approach is to use **Kruskal's algorithm**. In this method we:

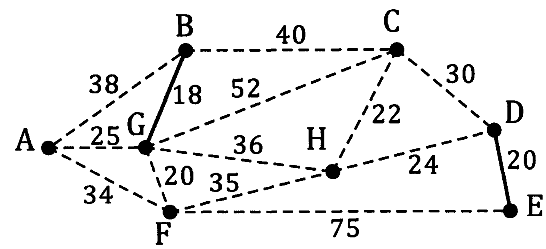
start with the "shortest" edge, then choose the next" shortest" edge, even if it is not connected to the previously selected edge, and then continue this process, always making sure that no cycles are introduced. When all vertices are "on line" we have the minimum spanning tree.

Using this Kruskal's approach for the network on the previous page:

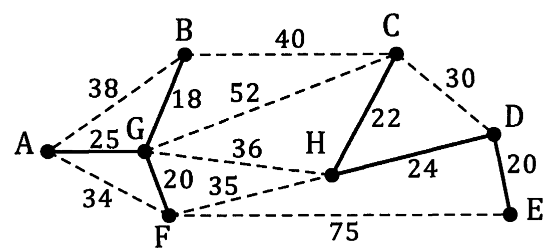
First we choose the "shortest" edge in the network, in this case GB which has a weighting of just 18.



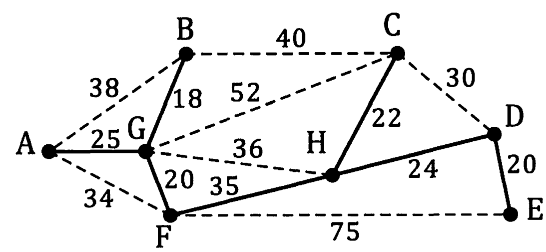
Then we choose the next "shortest" edge. In this case there are two, GF and DE, each with a weighting of 20. It does not matter which of these two we choose. We will choose DE, as shown.



Continuing this process sees us choose GF (20) then HC (22) then HD (24) and then AG (25) as our next four edges, as shown on the right.



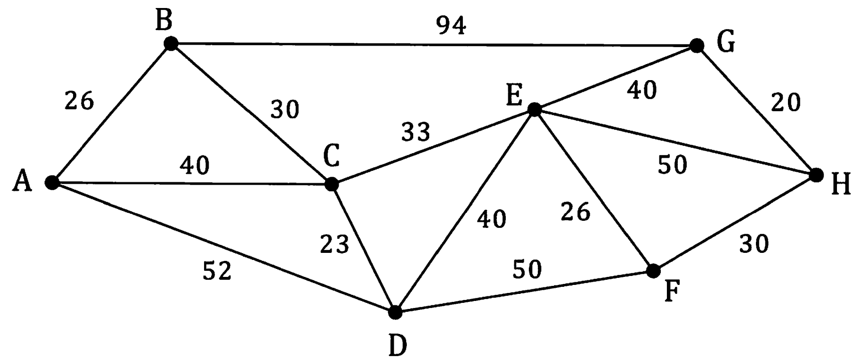
The next "shortest" edge will be CD (30). However this will not bring any vertex on line that is not already on line. (It will give an unnecessary cycle.)
The next "shortest" after CD is AF (34) which again introduces an unnecessary cycle.
Choosing the next "shortest" which does not introduce a cycle we choose FH (35), and so the minimum spanning tree is completed.



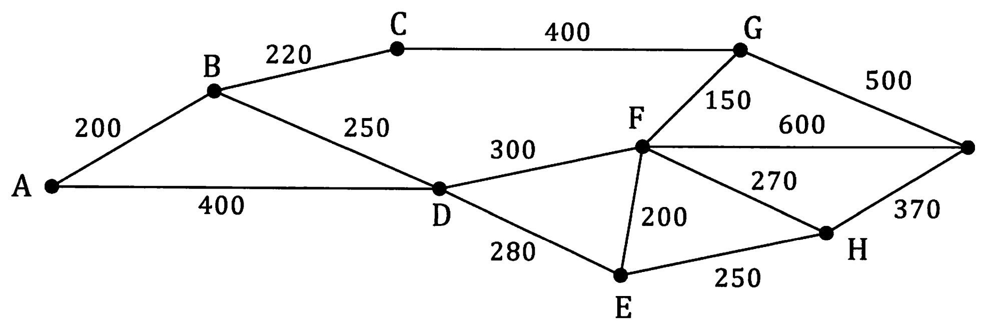
Exercise 5A

Find the minimum spanning tree for each of numbers 1 to 6, and state its length in each case.

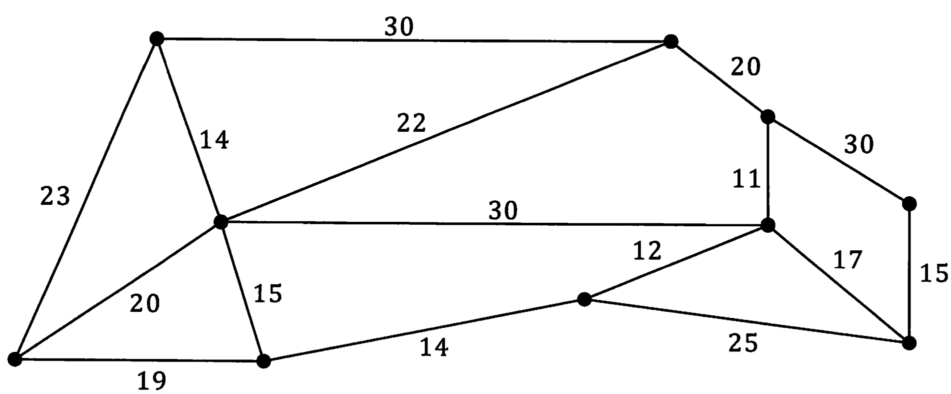
1.



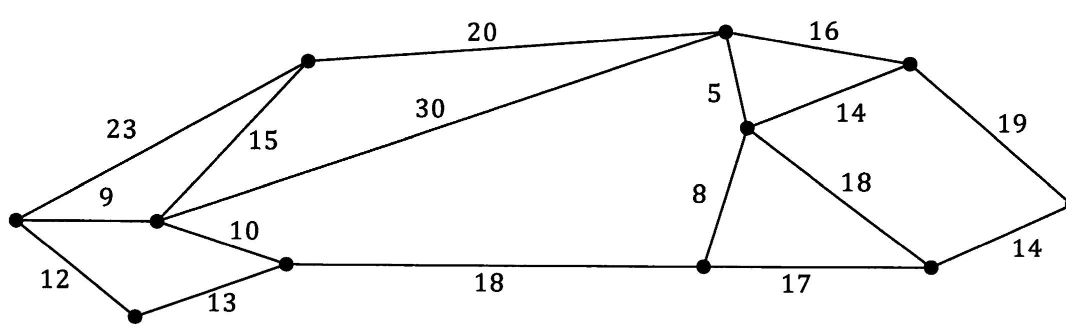
2.



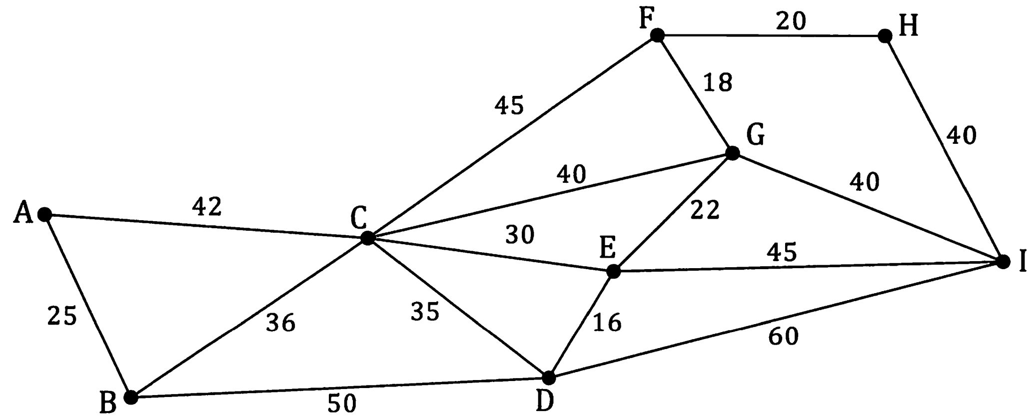
3.



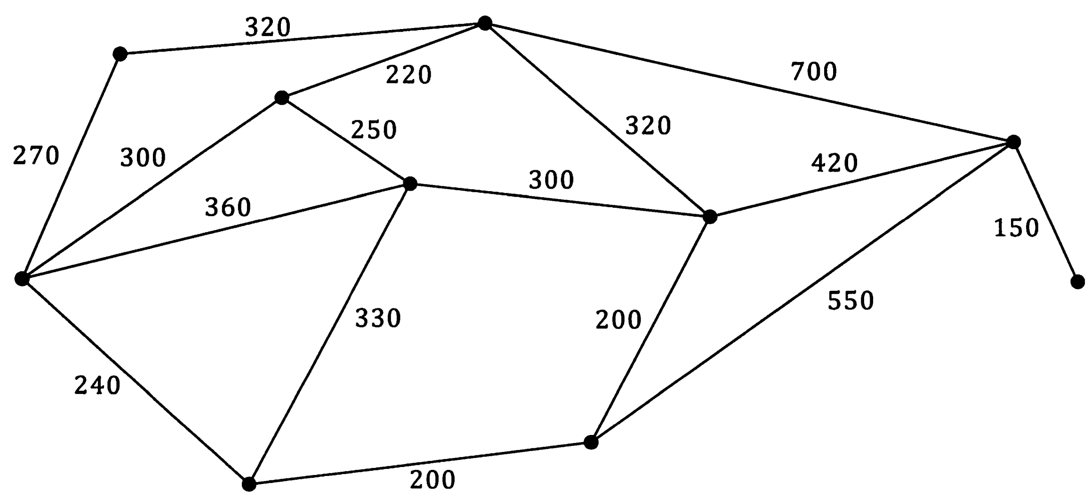
4.



5.



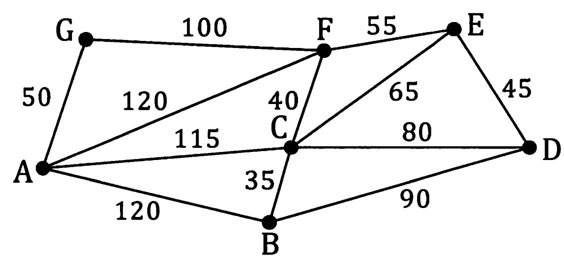
6.



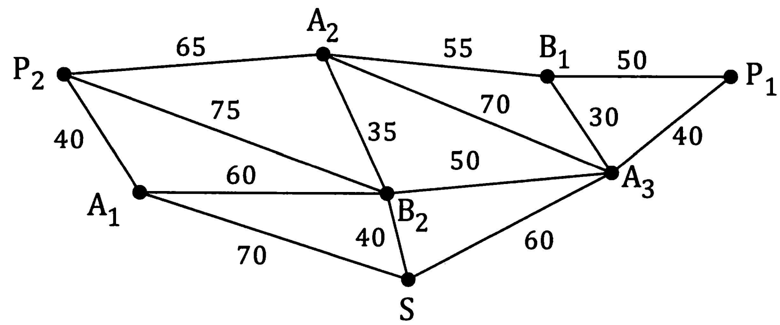
7. A market gardener wishes to replace the old piping currently connecting his seven outlet irrigation system. Rather than replacing all of the existing piping he wishes to replace the system with the minimum total length of piping that will keep all seven outlets connected, either directly or indirectly. He will then close off those unnecessary parts of the existing system.

The graph on the right shows the existing system. Points A to G are the outlets and the numbers indicate the length of each section in metres.

Which sections should he replace with new piping, which should he close off, and what total length of new piping will be needed?

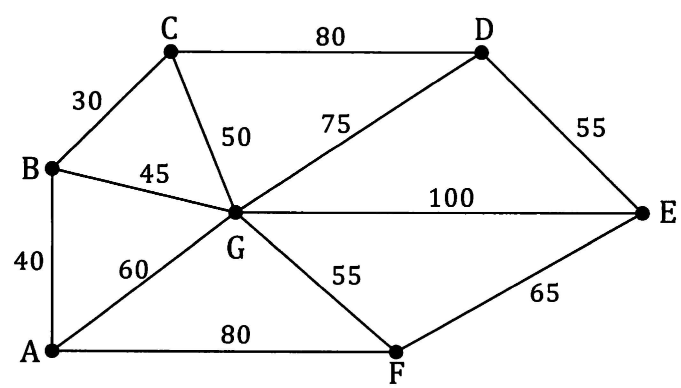


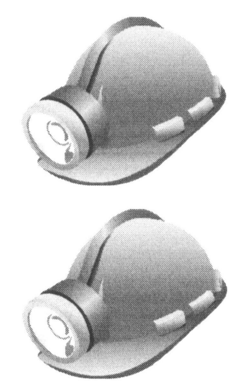
8. A camp-site is being designed with three ablution blocks, A_1, A_2 and A_3, two play areas for children, P_1 and P_2, two barbecue areas, B_1 and B_2, and a shop, S. The sketch below shows the relative locations of these features with the distances between them given in metres.



Paths are to be built so that all of these features are linked. On a copy of the above diagram indicate the network of paths that spans all of the features whilst minimising the total length of pathway.
What is this minimum length of pathway?

9. The network below shows the system of tunnels that exist linking the seven entrances of an old silver mine. The numbers on the edges show the length of each section of tunnel in metres.





The mine is a popular tourist attraction but all of the tunnels are in need of repair if they are to remain open for visitors.
The owners cannot afford to repair all of the tunnels so instead decide to repair the minimum total length possible, while still allowing entrance to the repaired system to be possible from all seven of the existing entrances.
The tunnels not being repaired will then be closed off to visitors.
They ask you to determine which tunnels should be repaired and which shut down.

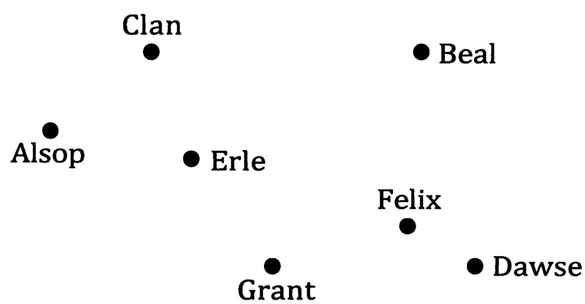
(a) On a copy of the network show the tunnels the owners should repair.

(b) What is the total length of tunneling that will be closed off to visitors?

10. The table on the right shows the distances, in km, along the roads that exist between seven towns:
Alsop, Beal, Clan, Dawse, Erle, Felix, and Grant.
(A "–" indicates that there is no direct road between those towns.)

	Alsop	Beal	Clan	Dawse	Erle	Felix	Grant
Alsop	–	–	310	–	300	–	620
Beal	–	–	550	430	530	330	–
Clan	310	550	–	–	350	–	–
Dawse	–	430	–	–	–	210	390
Erle	300	530	350	–	–	380	360
Felix	–	330	–	210	380	–	–
Grant	620	–	–	390	360	–	–

The sketch on the right shows the approximate positions of the seven towns with respect to each other.



(a) Make a copy of this sketch and show on your copy the roads that exist between the towns and the lengths of these roads.

(b) Water pipes are to be laid along some of these roads so that the seven towns are, either directly or indirectly, connected to each other.

Determine the minimum length of piping required.

Minimum spanning tree from the distances table.

In the last question of the previous exercise it was reasonably easy to transfer the details given in the table onto a network diagram because we were given a diagram showing the approximate locations of the towns with respect to each other. We then used the network diagram to determine the minimum spanning tree. However, if the network involved had been larger, and/or the relative locations of the vertices not have been known, the task of creating the network diagram with all of the weightings in place would have been more difficult. Fortunately it is possible to determine the minimum spanning tree directly from the table of distances without having to create the network diagram first. This process is demonstrated on the next page. However, as you read through it you should notice that the method is actually Prim's algorithm ....

choose any vertex as our initial "on line" vertex and then build up the spanning tree by connecting online vertices to the "nearest neighbour" whilst always making sure that no cycles are introduced

applied to the table of distances rather than to the network drawing.

Consider the table of distances shown on the right, which is actually the one from the last question of the previous exercise.

	A	B	C	D	E	F	G
A	–	–	310	–	300	–	620
B	–	–	550	430	530	330	–
C	310	550	–	–	350	–	–
D	–	430	–	–	–	210	390
E	300	530	350	–	–	380	360
F	–	330	–	210	380	–	–
G	620	–	–	390	360	–	–

Whilst we can start at any vertex we will choose to start at A.
With A *on line* we look down the A column to see which is the shortest connection that can be made from A.
In this case it is the 300 km connection to town E.

	A	B	C	D	E	F	G
A	–	–	310	–	300	–	620
B	–	–	550	430	530	330	–
C	310	550	–	–	350	–	–
D	–	430	–	–	–	210	390
E	300	530	350	–	–	380	360
F	–	330	–	210	380	–	–
G	620	–	–	390	360	–	–

Having chosen to look down the *A column* for a connection <u>from</u> A we then rule a line through the *A row* to ensure that we do not, at some later stage, use this line to make an unnecessary connection back <u>to</u> A.
(Alternatively, we could have chosen to look across *row* A for the shortest connection, and then rule a line through *column* A.)

	A	B	C	D	E	F	G
~~A~~			~~310~~		~~300~~		~~620~~
B	–	–	550	430	530	330	–
C	310	550	–	–	350	–	–
D	–	430	–	–	–	210	390
E	300	530	350	–	–	380	360
F	–	330	–	210	380	–	–
G	620	–	–	390	360	–	–

Having selected the 300 km connection to vertex E we now have both A and E *on line* and indicate this by the arrows at the top of these two columns.
Again, to avoid making unnecessary connections at a later stage we rule a line through the E *row*.
We now look down our two *on line* columns, A and E, and choose the minimum connection that can be made from one of these two vertices.

	A	B	C	D	E	F	G
~~A~~			~~310~~		~~300~~		~~620~~
B	–	–	550	430	530	330	–
C	310	550	–	–	350	–	–
D	–	430	–	–	–	210	390
~~E~~	~~300~~	~~530~~	~~350~~			~~380~~	~~360~~
F	–	330	–	210	380	–	–
G	620	–	–	390	360	–	–

This gives us the 310 km road from A to C as our next connection.

This brings vertex C on line and we rule a line through the C row.

	A	B	C	D	E	F	G
~~A~~			~~310~~		~~300~~		~~620~~
B	–	–	550	430	530	330	–
~~C~~	310	~~550~~			~~350~~		
D	–	430	–	–	–	210	390
~~E~~	300	~~530~~	~~350~~			~~380~~	~~360~~
F	–	330	–	210	380	–	–
G	620	–	–	390	360	–	–

We now look down our three *on line* columns, A, C and E, and choose the minimum connection that can be made from one of these three vertices.

This gives us the 360 km road from E to G as our next connection.

	A	B	C	D	E	F	G
~~A~~			~~310~~		~~300~~		~~620~~
B	–	–	550	430	530	330	–
~~C~~	310	~~550~~			~~350~~		
D	–	430	–	–	–	210	390
~~E~~	300	~~530~~	~~350~~			~~380~~	~~360~~
F	–	330	–	210	380	–	–
G	620	–	–	390	360	–	–

This brings town G on line and we rule a line through the G row.

We continue this process until all of the towns are on line. The circled items in the table indicate the direct connections that together form the minimum spanning tree.

	A	B	C	D	E	F	G
~~A~~			~~310~~		~~300~~		~~620~~
B	–	–	550	430	530	330	–
~~C~~	310	~~550~~			~~350~~		
D	–	430	–	–	–	210	390
~~E~~	300	~~530~~	~~350~~			~~380~~	~~360~~
F	–	330	–	210	380	–	–
~~G~~	~~620~~			~~390~~	360		

From the completed table on the right we see that the roads AC, AE, EF, EG, FB and FD form the minimum spanning tree, with a total length of 1890 km.
The reader should confirm that this is the same answer as was obtained for question 10 of Exercise 5A.

	A	B	C	D	E	F	G
~~A~~			~~310~~		~~300~~		~~620~~
~~B~~			~~550~~	~~430~~	~~530~~	330	
~~C~~	310	~~550~~			~~350~~		
~~D~~		~~430~~				210	390
~~E~~	300	~~530~~	~~350~~			~~380~~	~~360~~
~~F~~		~~330~~		~~210~~	380		
~~G~~	~~620~~			~~390~~	360		

This application of **Prim's algorithm**, may at first seem rather tedious and lengthy but you will find that it can be carried out quite quickly and, of course, in practice it only requires one copy of the table. The multiple copies shown above were to allow each step of the process to be clearly seen.

Exercise 5B

Determine the length of the minimum spanning trees for the networks defined by the tables in questions 1 to 5. (A "–" in the table indicates there is no direct route between the locations.)

1.

	A	B	C	D
A	–	–	21	20
B	–	–	39	35
C	21	39	–	24
D	20	35	24	–

2.

	A	B	C	D	E
A	–	38	–	62	32
B	38	–	37	–	35
C	–	37	–	40	22
D	62	–	40	–	41
E	32	35	22	41	–

3.

	A	B	C	D	E	F
A	–	450	–	–	–	470
B	450	–	300	–	–	400
C	–	300	–	500	200	400
D	–	–	500	–	420	700
E	–	–	200	420	–	310
F	470	400	400	700	310	–

4.

	A	B	C	D	E	F	G
A	–	2·6	–	–	–	3·6	4·0
B	2·6	–	5·0	–	–	–	2·8
C	–	5·0	–	5·0	3·0	–	3·5
D	–	–	5·0	–	5·3	–	–
E	–	–	3·0	5·3		5·7	3·9
F	3·6	–	–	–	5·7	–	2·8
G	4·0	2·8	3·5	–	3·9	2·8	–

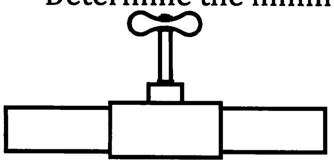
5.

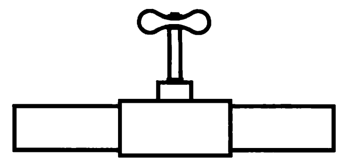
	A	B	C	D	E	F	G	H	I
A	–	650	–	–	–	370	400	–	–
B	650	–	600	–	–	–	340	350	–
C	–	600	–	220	–	–	–	–	380
D	–	–	220	–	560	–	–	–	390
E	–	–	–	560	–	700	–	300	210
F	370	–	–	–	700	–	–	410	–
G	400	340	–	–	–	–	–	–	–
H	–	350	–	–	300	410	–	–	250
I	–	–	380	390	210	–	–	250	–

6. Determine the length of the minimum spanning tree for the network defined by the table on the right. (A "–" in the table indicates there is no direct route between the locations.) Name the five roads that would be used in your minimum spanning tree.

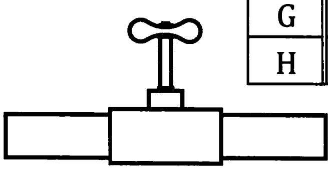
	A	B	C	D	E	F
A	–	8·7	–	8·9	–	9·1
B	8·7	–	9·5	–	–	4·5
C	–	9·5	–	6·1	7·0	5·3
D	8·9	–	6·1	–	3·5	–
E	–	–	7·0	3·5	–	3·8
F	9·1	4·5	5·3	–	3·8	–

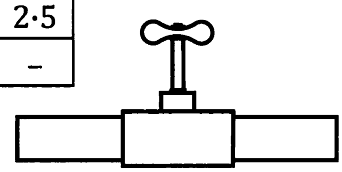
7. Eight farmlets (A, B, C, D, E, F, G and H) are to be connected to mains water supply. The water authority plans to lay pipes alongside some of the existing roads that link the farmlets. The plan is to bring all eight farmlets "on line" using those roads that form the minimum spanning tree for the network of existing roads. In this case "minimum" meaning the minimum total length. The lengths of the existing roads are as in the table below. All lengths are given in kilometres and a "–" in a space means that there is no existing road directly linking those two places.
Determine the minimum length of pipe required.





	A	B	C	D	E	F	G	H
A	–	4·2	–	–	7·2	4·1	–	–
B	4·2	–	6·0	–	–	2·0	–	–
C	–	6·0	–	4·4	–	6·3	3·0	–
D	–	–	4·4	–	6·8	–	5·0	6·0
E	7·2	–	–	6·8	–	–	–	2·3
F	4·1	2·0	6·3	–	–	–	3·5	2·8
G	–	–	3·0	5·0	–	3·5	–	2·5
H	–	–	–	6·0	2·3	2·8	2·5	–





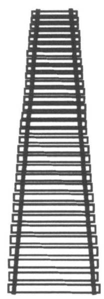
8. A school has six teaching areas, A, B, C, D, E and F, that need to be linked for the school video, public address, internet and phone systems.

Cables are to be installed to create a spanning tree for the network with vertices at A, B, C, D, E and F.

Find the minimum cost of the spanning tree if the cost of the direct cable links between vertices are as in the table on the right.

	A	B	C	D	E	F
A	0	$170	$525	$410	$210	$325
B	$170	0	$400	$310	$145	$225
C	$525	$400	0	$140	$315	$200
D	$410	$310	$140	0	$190	$100
E	$210	$145	$315	$190	0	$115
F	$325	$225	$200	$100	$115	0

9. A railway authority wishes to build a freight railway network linking seven towns. The network is to be constructed so that it will be possible to travel by rail from any one of the towns to any of the other six in the network, perhaps not directly, but at least by going via other towns in the network. The towns are Harlow, Gatley, Olber, Wayley, Raine, Lewis and Fitch and the distances involved for the feasible rail routes from any of these towns to any of the others are given in the following table (in kilometres).



	Harlow	Gatley	Olber	Wayley	Raine	Lewis	Fitch
Harlow	0	300	510	316	122	195	320
Gatley	300	0	224	100	212	378	206
Olber	510	224	0	200	400	517	269
Wayley	316	100	200	0	202	327	112
Raine	122	212	400	202	0	189	198
Lewis	195	378	517	327	189	0	252
Fitch	320	206	269	112	198	252	0

(a) Based on these distances determine the minimum length of track required for this network.

(b) In this "minimum length network" which six pairs of towns would have a direct rail link? (By direct rail link we mean able to journey from one town to the other without having to pass through any of the other towns on the way.)

10. The table on the right indicates the distances, in kilometres, along roads that exist between ten locations, A to J. (A shaded cell in the table indicates that there is no direct road between the locations.)

	A	B	C	D	E	F	G	H	I	J
A				60	30		70			
B					45	20				40
C				65				28	35	30
D	60		65		46	18	20	43		
E	30	45		46		42				
F		20		18	42					41
G	70			20				38		
H			28	43			38		54	
I			35					54		
J		40	30			41				

The roads are in need of upgrading and it is decided that, rather than upgrade all of the roads, only those roads needed to ensure that all ten locations are linked by upgraded roads, either directly or indirectly, will be upgraded.

Which roads should be upgraded if the total length of road to be upgraded is to be kept to a minimum and what would this minimum length be?

Miscellaneous Exercise Five.

This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

1. The following employment data gives the number of people in full time employment in Australia each quarter from January 2010 to April 2013, with the exception of July 2011 for which the information is missing from the table.

Data point (n)	1	2	3	4	5	6	7
Month	Jan 2010	April 2010	July 2010	Oct 2010	Jan 2011	April 2011	July 2011
Employed ('000) (N)	7630	7677	7749	7821	7876	7897	

Data point (n)	8	9	10	11	12	13	14
Month	Oct 2011	Jan 2012	April 2012	July 2012	Oct 2012	Jan 2013	April 2013
Employed ('000) (N)	7911	7942	7971	7989	8024	8028	8022

(Based on Australian Bureau of Statistics data.)

(a) Use linear regression techniques to determine the equation of the least squares line of best fit, N = an + b.

(b) Interpret your value of a in your answer for part (a).

(c) Use your regression line to predict the employment figure for July 2011 and for January 2015 stating which answer you would expect to be the more reliable and why.

(d) View the given data on a graphic calculator or computer and write some appropriate comments.

2. In an attempt to determine how long an investment of $350000 placed in an account paying 6% per annum, compounded monthly, would last if $1500 was withdrawn after one month, and at the end of every month thereafter, Jim put this information into a financial calculator, as shown below left. When he asked the calculator to determine N, the number of withdrawals the calculator returned "Error", as shown below right. Explain.

Compound Interest	
N	
I%	6
PV	-350000
PMT	1500
FV	0
P/Y	12
C/Y	12

Compound Interest	
N	Error
I%	6
PV	-350000
PMT	1500
FV	0
P/Y	12
C/Y	12

3. The table below shows some of the data for the number of kilolitres of water used by a household each quarter for a period of three years.

t	Year	Quarter	N^o.^ of kL	Uncentred 4pt MA	Centred 4pt MA
1	One	1st	95		—
				—	
2	One	2nd	60		—
				a	
3	One	3rd	36		d
				b	
4	One	4th	77		78
				c	
5	Two	1st	131		e
				82	
6	Two	2nd	f		81
				g	
7	Two	3rd	44		79
				h	
8	Two	4th	i		k
				j	
9	Three	1st	123		m
				l	
10	Three	2nd	92		84
				n	
11	Three	3rd	60		—
				—	
12	Three	4th	o		—

Calculate the value of a, b, c, d, ... o.

4. To the nearest cent, what "one-off" lump-sum payment must be invested into an account paying compound interest of 8% per annum, compounded monthly, for the investment to be worth $5000 four years later?

5. Amy takes out a loan of $15000. Interest of 8·2% of the outstanding balance is added at the end of each year and then Amy's regular annual repayment of $2500 is taken from the outstanding balance.
   (a) How much does Amy still owe on this loan immediately after she makes the $2500 repayment at the end of year six?
   (b) If instead Amy wishes to pay the loan off in the six years what does her regular annual repayment need to be?